

HEMOGLOBIN HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY (HPLC) & BETA-GLOBIN SCREEN

|                     |                                               |                    |                                             |
|---------------------|-----------------------------------------------|--------------------|---------------------------------------------|
| Patient Name:       | Manpreet Kaur                                 | Date of Exam:      | September 14, 2026                          |
| MRN / Patient ID:   | CMC-THAL-2026-3399                            | Age / Gender:      | 6 Yrs / Female                              |
| Blood Group & Rh:   | O+ (Rh Positive)                              | Referring Doctor:  | Dr. Joseph John, MD (Clinical Hematology)   |
| Clinical Diagnosis: | Beta Thalassemia Major (Transplant Candidate) | Specimen / Source: | Whole Blood EDTA (Pre-Transfusion Baseline) |

| Hemoglobin Fraction     | Observed Peak % | Reference Range | Retention Time | Clinical Implication              |
|-------------------------|-----------------|-----------------|----------------|-----------------------------------|
| Hb F (Fetal Hemoglobin) | 88.4            | < 1.0           | 1.12 min       | Marked Elevation (Major)          |
| Hb A (Adult Hemoglobin) | 4.2             | 95.0 - 98.0     | 2.48 min       | Severe Suppression of Beta Chains |
| Hb A2 (Delta Chains)    | 5.6             | 1.8 - 3.5       | 3.62 min       | Elevated (Beta Thalassemia Trait) |
| HBB Gene Mutation       | IVS1-5 (G>C)    | Homozygous      | Sanger Seq     | Severe Beta-Zero Allele           |
| Serum Ferritin          | 1,840           | 15 - 150        | ng/mL          | Significant Iron Hemosiderosis    |
| Total Serum Bilirubin   | 3.1             | 0.2 - 1.2       | mg/dL          | Chronic Extravascular Hemolysis   |

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- Bio-Rad Variant II HPLC profile demonstrates severe Beta Thalassemia Major phenotype with HbF 88.4%.
- HBB DNA sequencing confirms homozygous IVS1-5 (G>C) mutation (most common severe Indian mutation).
- Elevated serum ferritin (1,840 ng/mL) requires pre-transplant oral iron chelation with Deferasirox.
- Recommendation: Lucarelli Pesaro Risk Class II; eligible for allogeneic matched sibling / MUD BMT.

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